JÖNKÖPING UNIVERSITY
School of Engineering

Industrial espionage

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TODAY

- Industrial espionage
- Information as an asset
- The effects of industrial espionage
- Some examples
- How to avoid it
- Phishing and spear phishing

LITTERATURE NOTE!

- Some of the lecture content is not covered by the book but may still appear on the exam
- The law stuff in the book is not written from a Swedish perspective, but I have integrated Swedish perspectives into the slides.

CLASSIFYING CRIME

- ”Private” crimes
  - Fraud, drugs, stalking, sex offenses etc
- **Crimes against companies**
  - **Industrial espionage**
- Crimes against states
  - Cyber warfare

INDUSTRIAL ESPIONAGE!

- The James Bond of cybercrime?
- All actions seeking to steal information from private companies
- Perpetrators are often adversarial states or competing companies

VALUE OF INFORMATION

- We live in the age of information

- Information assets are the new oil
  - Patent
  - Trade secrets
  - RnD
  - …

- Companies spend tons to research those types of stuff

VALUE OF INFORMATION

- Developing it
  - Research
  - Design

- What it can make you earn
  - Production and retail
  - Having the best products gives you an edge

VALUE OF INFORMATION

- Consider Tesla who was “first” with electric cars

- Costs
  - Years of development
  - Marketing

- Gains
  - The only option for several years

HAZARDS

- Espionage can obviously lead to someone else getting access to our data
- It can also mean someone gets a foothold in our systems
- Or discloses data to cause us harm

LETS PLAY A GAME

- Be the security chief of Fugle Co
- Make decisions on a budget and see if you can secure a successful app launch
- http://targetedattacks.trendmicro.com/index.html
  - Or google trend micro game (and go for Targeted attack: the game)
- Play and have a break :)

Reflect:1-2-4-all

Individually (60 Sec): Write core statements (one-liners) about what you learned in the last lecture part.

Pair up and compare lists (60 Sec) (add new things to your individual lists)

Pair the pairs and repeat (60 Sec)

Then one statement per group of four.

Questions?

EXAMPLE 1: RUSSIAN SPY IN SWEDEN

- P3 dokumentär (In Swedish)
  - Putins svenske agent och spionskandalen på Scania
- A Russian intelligence officer recruited a tech worker (the spy) at Volvo
- The spy photographed data with a camera and sold to the officer
- Ths spy then started working at Scania and did the same thing

EXAMPLE 1: RUSSIAN SPY IN SWEDEN

- The Officer
  - Was in sweden on a diplomatic passport and officially worked for the russian embassy
  - Got in contact with ”the spy” through mutual friends and formed a friendship
  - From there, recruited the spy

- The spy
  - Can be assumed to have been disgruntled and on the lookput for easy money

EXAMPLE 2: CHINESE ACTOR TARGETS US AND EUROPE COMPANIES

- In 2014 Solarworld was a world leading company in solar panel production
- They had data stolen from them in a large attack attributed to APT1 – a Chinese threat actor
- Allegedly, the intellectual property has since been used by Chinese companies

EXAMPLE 2 CONT

- Because of the attack, SolarWorld lost their competitive advantage with financial loss as a result

- Filed for bankruptcy in 2017 and reorganized, but filed for bankruptcy again in 2018.

EXAMPLE 2: CHINESE ACTOR TARGETS US AND EUROPE COMPANIES

- AP1 is known for IP theft/industrial espionage targeting western world companies

- Modus operandi
  - Spear phishing
  - Deployment of custom evil code
  - Data exfiltration

- Great report by Mandiant: https://www.mandiant.com/resources/reports/apt1-exposing-one-chinas-cyber-espionage-units

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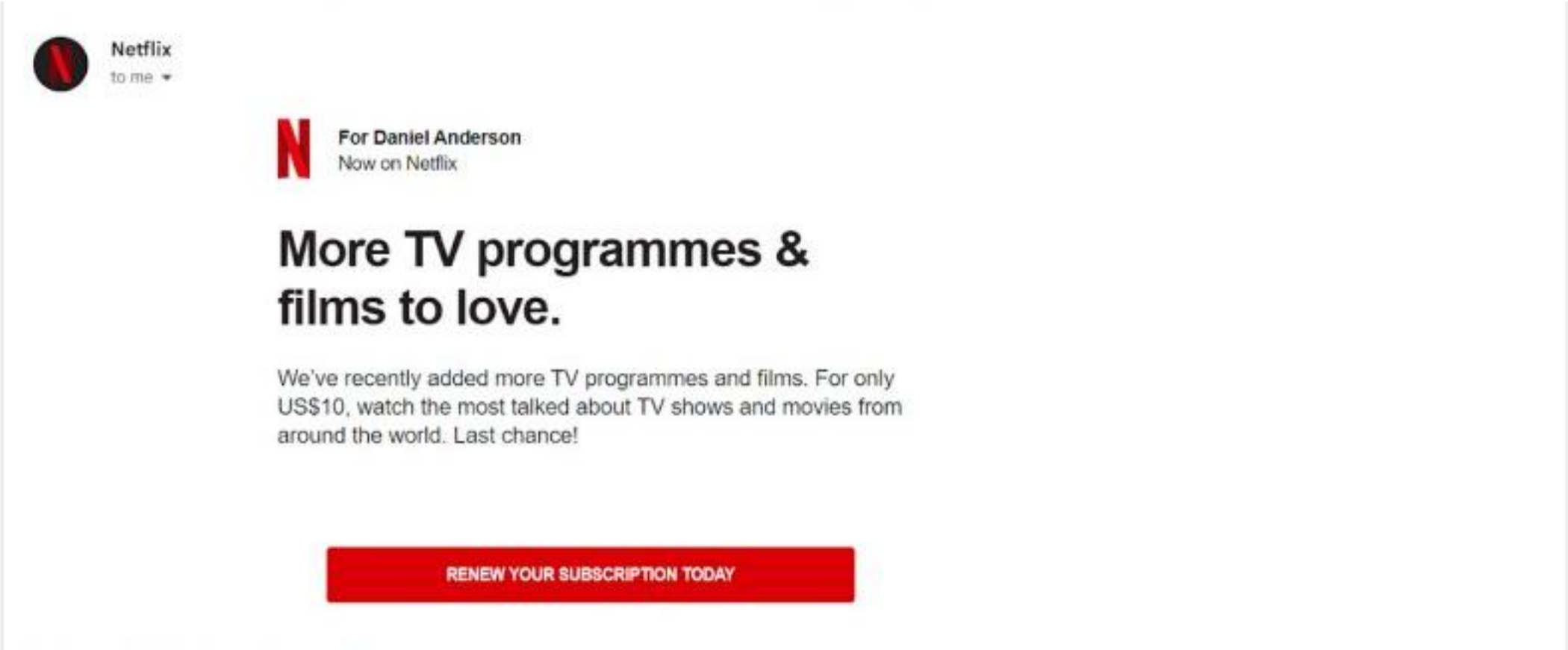
PHISHING?

- The act of tricking users to do things they shouldn't using email
  - Handing out login credentials
  - Hand out sensitive data
  - Click stuff, which then runs malware

- Phishing is available on multiple platforms
  - Smishing: via sms
  - Vishing: via phone calls
  - ….

PHISHING?


Netflix
to me
For Daniel Anderson
Now on Netflix
More TV programmes & films to love.
We've recently added more TV programmes and films. For only US$10, watch the most talked about TV shows and movies from around the world. Last chance!
RENEW YOUR SUBSCRIPTION TODAY

Reflect:1-2-4-all

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PHISHING?

Google

Gmail

Important: Your Password will expire in 1 day(s) Inbox x

MyUniversity 12:18 PM (50 minutes ago)

to me

Dear network user,

This email is meant to inform you that your MyUniversity network password will expire in 24 hours.
Please follow the link below to update your password
myuniversity.edu/renewal

MY UNIVERSITY

Thank you
MyUniversity Network Security Staff

WHY DOES IT WORK?

- The general idea is to trick people by exploiting fundamental human characteristics
  - Trust
  - Wanting to help
  - Scare or problems

- And exploit ways to make us do things without thinking
  - Sense of urgency
  - Sense of authority

WHY DOES IT WORK?

- It is also a numbers game
  - Sending 100 000 000 emails is cheap and easy
- The e-mail will appear relevant for some of those people
- And they might just click and get owned

SPEAR PHISHING?

- Same as phishing but with an email customized for one recipient, or a group of recipients.
- With some recon, it can be very easy to craft a good email.

1-2-4-all – phishing edition

Individually (60 Sec): Contemplate how you would best spear phish me☺.

Pair up (60 Sec) can you improve together?

Pair the pairs and repeat (60 Sec)

What did you come up with?

PHISHING – STAY SAFE

NEVER EVER OPEN FILES YOU DONT EXPECT

PHISHING – STAY SAFE

NEVER EVE
HAND OUT
INFORMATION
WHEN ASKED

Including passwords
and usernames!

PHISHING – STAY SAFE

No, really!

PHISHING – STAY SAFE

- Also, very that the sender email is true
  - Typically by hovering over the sender name

- And that the link address is correct
  - Typically by hovering over the link

SOURCES

- Course book
- Some optional scientific papers:
  - Kävrestad, J., & Huskaj, G. (2021). How the Civilian Sector in Sweden Perceive Threats From Offensive Cyberspace Operations. *20th European Conference on Cyber Warfare and Security, Reading, 24-25 June 2021.*
  - Kävrestad, J., Hagberg, A., Nohlberg, M., Rambusch, J., Roos, R., & Furnell, S. (2022). Evaluation of Contextual and Game-Based Training for Phishing Detection. *Future Internet*, *14*(4), 104.
  - Kävrestad, J., Pettersson, R., & Nohlberg, M. (2020). The language effect in phishing susceptibility. *6th International Workshop on Socio-Technical Perspective in IS Development (STPIS'20), Virtual Conference in Grenoble, France, June 8-9, 2020*, 162–167.

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